Faculty Guide · Lecture Videos & Simulations · Modules 01–03

PCCN Certificate Course · IAP Intensive Care Chapter · 15–29 October 2026

For faculty recording the **13 pending lecture videos** and building the **4 simulation scenarios** in Modules 01–03. Everything here is sorted into four categories so you know what is non-negotiable and what is optional.

Must follow

Production rules for every video.

Must cover

Clinical content each video has to teach. The module quiz is written from these points.

Better to follow

Recommended teaching style. Strongly encouraged, not checked.

Good to know

Optional pearls if time allows. Label them "beyond the exam".

1. Must follow — production rules

1. **Length:** stay within the cap listed for your video (±2 min). If it runs longer, split it into two parts rather than speeding up.

2. **Structure:** use the fixed running order in section 2.

3. **Template:** PCCN slide template only. Title card shows the module number; end card shows "Next: quiz / simulation".

4. **Format:** 1080p, 16:9 landscape, MP4. Quiet room, external or headset mic, no echo or fan noise.

5. **Patient privacy:** no names, faces, bed numbers or hospital IDs. Any clinical photo or clip needs written consent and blurred faces.

6. **Evidence:** follow current AHA PALS and IAP guidance. Give drug doses per kg with the maximum dose, and use generic drug names only drug dosages should also have standard references.

7. **No commercial content:** no pharma or device logos, and no product endorsements.

8. **Language:** British spelling and SI units on slides; speak to the bedside nurse ("you").

9. **Quiz alignment:** every Must cover point is taught clearly enough that the module quiz can be answered from the video.

10. **Delivery:** upload as YouTube *Unlisted* (or Google Drive "Anyone with the link"). File name format: PCCN_M01_V2_PAT.

2. Must follow — running order of every video

Time	Segment
0:00–0:15	Title card: PCCN · Module · number · Title · Faculty name
0:15–1:00	Bedside case hook: one child, one problem, one question
1:00–1:30	Learning objectives (3 minimum)
Core	2–4 teaching chunks. Each chunk ends with a concrete **nurse action**.
Last 1:30	Return to the case → 3 take-home points → pointer to the quiz or simulation
0:10	End card

3. Better to follow — teaching style

• Picture-in-picture: your face in a corner over the slides keeps nurses watching.

• One idea per slide, 6 lines maximum. Reveal mnemonics letter by letter.

• Demonstrate with real equipment used in Indian PICUs rather than describing it.

• Add one "pause and decide" prompt in the core, then give the answer.

• Focus on what the nurse does, sees and escalates, not physician decisions.

• Provide an English subtitle file (.srt) if possible.

4. Video-by-video brief

Must cover points are required. **Good to know** points are optional.

Module 01 · Initial Assessment

V1 Why Children Are Different ≤10 min · record by Mon 19 Oct

Must cover

• Airway: large occiput, large tongue, narrow subglottis, infants breathe through the nose

• Low respiratory reserve: high O_2 use, soft chest wall, tires quickly

• Cardiac output depends on heart rate; low blood pressure is a *late* sign

• Weight-based dosing and how to estimate weight

• Heat and glucose loss; family-centred care

Good to know: length-based tape; adjusting your approach to the child's developmental stage.

V2 PAT & Primary Assessment ≤16 min · record by Mon 19 Oct

Must cover

• The 3 sides of the PAT: Appearance (TICLS), Work of Breathing, Circulation to Skin

• The PAT is done from the doorway in 30–60 seconds, without touching the child

• From PAT to first impression: stable, respiratory distress or failure, shock, CNS dysfunction, cardiopulmonary failure

• ABCDE hands-on assessment with normal vital-sign ranges by age

• Findings that need immediate escalation

Good to know: SAMPLE history for the secondary assessment; preview of PEWS.

V3 The DIRECT Approach ≤14 min · record by Tue 20 Oct

Must cover

• D Detection · I Intervention · R Reassessment · EC Effective Communication · T Transport, with nurse actions at each step

• Reassess after every intervention and at least every 5 minutes while the child is unstable

• ISBAR handover and closed-loop communication

• Checklist for safe transfer within the hospital

Good to know: overlap with crisis resource management; documenting during a resuscitation.

Module 02 · Respiratory Support

V4 Respiratory Distress vs Failure ≤10 min · record by Wed 21 Oct

Must cover

• Signs of distress: fast breathing, recessions, nasal flaring, grunting, head bobbing

• Signs of failure: exhaustion, slowing breathing, silent chest, cyanosis, drowsiness

• The 3 E's: Effort, Efficacy, Effects

• Types of problem: upper airway, lower airway, lung tissue, disordered control of breathing

• When to call for help

Good to know: basic blood-gas patterns in respiratory failure.

V5 Oxygen Delivery Systems ≤12 min · record by Wed 21 Oct

Must cover

• Low-flow vs high-flow devices: nasal prongs, simple mask, non-rebreather mask, with flow rates and approximate FiO_2

• SpO_2 target 94–98%, unless the child's condition sets a different target

• Humidification, and choosing a device by the child's size and tolerance

• Escalation ladder and how to wean

Good to know: how to calculate how long an oxygen cylinder will last; oxygen risks in preterm infants.

V6 Bag-Mask Ventilation ≤14 min · record by Thu 22 Oct

Must cover

• When to start; choosing the bag size, and a mask that covers both nose and mouth

• Head position (neutral for infants, sniffing for older children); C-E grip; two-person technique

• Rate as per current PALS; watch for chest rise; avoid over-ventilating the child or inflating the stomach

• Troubleshooting when the chest does not rise

Good to know: PEEP valve; sizing an oropharyngeal airway.

V7 High-Flow Nasal Cannula (HFNC) ≤14 min · record by Thu 22 Oct

Must cover

• How it works: flushes dead space, gives a PEEP-like effect, delivers heated humidified gas

• When it is used, how to set it up, and starting flow per your unit protocol

• Signs it is working within 1–2 hours (HR, RR, SpO_2) and signs it is failing

• Nursing care: prongs fill less than half of each nostril, skin checks, feeding, positioning

Good to know: ROX index; using HFNC for apnoeic oxygenation.

V8 The Art of Nebulisation ≤10 min · record by Fri 23 Oct

Must cover

• Common drugs and when to use them: salbutamol, ipratropium, adrenaline, hypertonic saline

• Jet nebuliser vs MDI with spacer; oxygen-driven nebuliser at 6–8 L/min

• Technique: mask fit and position; a crying child receives less of the drug

• Monitoring side effects; cleaning and infection control

Good to know: continuous nebulisation; spacer-and-mask technique in infants.

Module 03 · Shock & Haemodynamics

V9 Clinical Features of Dehydration ≤10 min · record by Fri 23 Oct

Must cover

• Grading severity: no, some, or severe dehydration

• Bedside signs: alertness, eyes, tears, mucosa, fontanelle, skin pinch, CRT, urine output

• ORS Plans A, B and C, and when to switch to IV

• Intake–output charting and weighing

Good to know: dehydration with high or low sodium.

V10 Types of Shock — Type and Severity ≤12 min · record by Sat 24 Oct

Must cover

• Hypovolaemic, distributive (septic, anaphylactic), cardiogenic, obstructive

• Compensated vs hypotensive shock; hypotension = SBP below 70 + (2 × age in years)

• Signs: heart rate, CRT, central vs peripheral pulses, mental status, urine output

• Warm vs cold shock at the bedside

Good to know: using lactate trends.

V11 Fluid Resuscitation ≤14 min · record by Sat 24 Oct

Must cover

• IV or IO access within 5 minutes; check glucose

• Bolus of 10–20 mL/kg isotonic crystalloid (smaller and slower in cardiogenic shock); push–pull technique

• Reassess after every bolus; signs of fluid overload (crackles, enlarging liver, rising RR)

• Fluid-refractory shock: when to prepare vasoactive drugs

Good to know: balanced crystalloids vs normal saline; when blood products are needed.

V12 NIBP — How to Choose ≤8 min · record by Sun 25 Oct

Must cover

• Cuff size: bladder width about 40% of arm circumference, length 80–100%

• Cuff site and limb position; a cuff too small over-reads, too large under-reads

• How often to measure in shock; trending MAP

• When to move to invasive BP monitoring

Good to know: using Doppler for BP in small infants.

V13 The Physiologically Difficult Airway ≤15 min · record by Sun 25 Oct

Must cover

• The physiological dangers of intubation: hypoxaemia, hypotension, acidosis, right-heart strain

• Resuscitate before you intubate: fluids, push-dose adrenaline ready, preoxygenation

• SOAPME checklist and the nurse's role during intubation

• Watching for hypotension after intubation

Good to know: induction drugs that protect the haemodynamics; apnoeic oxygenation.

5. Simulations — 4 scenarios

Must follow — every scenario

• 10–15 minutes long. Case stem → monitor showing vitals that change → 3–5 decision points → feedback on every choice → debrief questions.

• Every wrong choice shows its consequence on the monitor and a teaching line.

• Uses the DIRECT approach and the same vital ranges taught in the videos.

• Faculty provide the storyboard: one row per decision, listing options, the correct answer, the reason, and how the vitals change.

Scenario	Must cover — decision points
S1 · M01 Virtual Triage A 8-month-old, respiratory distress	PAT impression → A/B checks → oxygen → triage as urgent → ISBAR call
S2 · M01 Virtual Triage B 4-year-old, minor injury	PAT stable → focused assessment → triage down safely → advice to family
S3 · M02 Escalation Drill 3-year-old, worsening bronchiolitis	Distress vs failure → choose device → escalate to HFNC → bag-mask when apnoeic → ISBAR handover
S4 · M03 The Septic Child 6-year-old, fever, low BP, slow CRT	PAT → ABCDE → classify shock → IV/IO + bolus → reassess → fluid-refractory → prepare vasoactive drug

Better to follow: use a realistic parent voice in the case stem and real drug-chart formats. **Good to know:** an optional "expert path" timer for faster learners.

6. 15-day timeline

Day	Milestone
D0 · Thu 15 Oct	Guide sent to faculty · videos and scenarios assigned · 30-min briefing call · slide template shared
D1 · Fri 16 Oct	Module 01 scripts and slide outlines (V1–V3) due
D2 · Sat 17 Oct	Module 02–03 scripts (V4–V13) due · all 4 simulation storyboards due
D3 · Sun 18 Oct	Course director reviews and signs off scripts and storyboards, with feedback within 24 h
D4–D5 · Mon 19–Tue 20	Record Module 01 (V1–V3) · start building S1–S2
D6–D8 · Wed 21–Fri 23	Record Module 02 (V4–V8) · M01 edited and reviewed · S1–S2 clinical review on Fri 23 · start building S3
D8–D10 · Fri 23–Sun 25	Record Module 03 (V9–V13) · M01 links wired into the portal · S1–S2 live on Sun 25 · start building S4
D11 · Mon 26 Oct	M02 edits approved and links wired · S3–S4 clinical review
D12 · Tue 27 Oct	M03 edits approved and links wired · S3–S4 live · **content freeze**
D13 · Wed 28 Oct	Full QA on mobile and desktop · pilot with 3 nurses · fix issues
D14 · Thu 29 Oct	**Go-live:** Modules 01–03 complete · announcement on WhatsApp

7. Task-by-task schedule

Task	Faculty	Script	Record / build	Edit & review	Live
V1 Why Children Are Different		16 Oct	19 Oct	21 Oct	**25 Oct**
V2 PAT & Primary Assessment		16 Oct	19 Oct	21 Oct	**25 Oct**
V3 The DIRECT Approach		16 Oct	20 Oct	22 Oct	**25 Oct**
V4 Distress vs Failure		17 Oct	21 Oct	23 Oct	**26 Oct**
V5 Oxygen Delivery Systems		17 Oct	21 Oct	23 Oct	**26 Oct**
V6 Bag-Mask Ventilation		17 Oct	22 Oct	24 Oct	**26 Oct**
V7 HFNC		17 Oct	22 Oct	24 Oct	**26 Oct**
V8 The Art of Nebulisation		17 Oct	23 Oct	25 Oct	**26 Oct**
V9 Clinical Features of Dehydration		17 Oct	23 Oct	25 Oct	**27 Oct**
V10 Types of Shock		17 Oct	24 Oct	26 Oct	**27 Oct**
V11 Fluid Resuscitation		17 Oct	24 Oct	26 Oct	**27 Oct**
V12 NIBP — How to Choose		17 Oct	25 Oct	26 Oct	**27 Oct**
V13 Physiologically Difficult Airway		17 Oct	25 Oct	26 Oct	**27 Oct**
S1–S2 M01 Virtual Triage A + B		17 Oct	19–22 Oct	23 Oct	**25 Oct**
S3 M02 Escalation Drill		17 Oct	21–24 Oct	26 Oct	**27 Oct**
S4 M03 The Septic Child		17 Oct	23–25 Oct	26 Oct	**27 Oct**

8. To-do list by owner

Course director

☐ Assign a faculty member to each of V1–V13 and S1–S4, and fill in the Faculty column (15 Oct)

☐ Run the briefing call and share this guide and the slide template (15 Oct)

☐ Review and sign off all scripts and storyboards (18 Oct)

☐ Clinical review of each edited video within 48 h of receiving it

☐ Clinical review of S1–S2 (23 Oct) and S3–S4 (26 Oct)

☐ Approve content freeze (27 Oct) and go-live (29 Oct)

Each faculty member

☐ Submit script and slide outline that cover every Must cover point (16 or 17 Oct)

☐ Simulation faculty: submit a storyboard with decision points, options, answers and vitals (17 Oct)

☐ Record within the length cap on your record-by date

☐ Run the pre-upload checklist below, then upload as Unlisted and send the link

☐ Make any review corrections within 24 h

Portal / tech

☐ Wire approved video links into Module 01 (25 Oct), 02 (26 Oct) and 03 (27 Oct)

☐ Build the S1–S4 interactive simulations from the storyboards

☐ QA on phone and desktop; check progress saves to the admin dashboard (28 Oct)

☐ Upload the final module files to GitHub Pages; post the WhatsApp announcement (29 Oct)

Pre-upload checklist (faculty)

☐ Within the length cap

☐ Title card and end card

☐ Case hook + 3 objectives

☐ Every Must cover point taught

☐ Recap with 3 take-home points

☐ No patient identifiers

☐ Doses per kg + maximum, generic names

☐ No logos or brands

☐ 1080p, clear audio

☐ File named PCCN_M0X_VX_Title